

cent change in X . Thus, if Y represents the quantity of a commodity demanded and X its unit price, then β_1 measures the price elasticity of demand.

The log-linear regression model has certain advantages: (i) the parameters are invariant to change of scale, since they measure percentage changes, (ii) the model gives elasticity figures directly, and (iii) the model moderates the problem of heteroscedasticity to some extent (see Unit 11 for the problem of heteroscedasticity).

1.6 SOFTWARE PACKAGES FOR ECONOMETRIC ANALYSIS

As econometric analysis deals with empirical data it involves cumbersome computations. We can think of estimating the parameters in some of the simpler estimation methods (such as OLS) involving a small dataset. In other cases, however, it is quite difficult and time consuming.

In recent years, easy access to computers and the availability of econometric software packages has made the job simpler. We find that the most difficult computations can be done by computers in a few seconds. There are several software packages available for econometric applications. Most of these software packages include the important econometric applications. Further, since these software packages apply similar methods, the results are almost the same. Some of these software are freely available while others need to be bought from the market. An open-source software package (i.e., freely downloadable from the Internet) is R. You can modify this software according to your need share with others without any licensing violation. Thus, a lot of new features (i.e., new methods and test statistics) are being added to this software. A major limitation of R is that you need to have some basic knowledge of computer programming.

Software such as STATA, E-Views and SPSS are more user-friendly in the sense that you can apply various econometric methods by the click of a mouse – you do not have to write programs. These software, however, are licensed and need to be purchased. There are certain freeware packages such as Gretl, EasyReg, Matrixer, etc. which can be freely downloaded from the Internet and can be used for econometric analysis.

Check Your Progress 2

- 1) Explain the need for inclusion of an error variable in a regression model.
.....
.....
.....
- 2) Explain the concept of data generation process. Is it the same as the estimated regression model?
.....
.....

3) How do you interpret the slope parameter of the following regression models?

(i) $Y_i = \beta_0 + \beta_1 X_i$

(ii) $\ln Y_t = \beta_0 + \beta_1 t + u_t$

(iii) $\ln Y_i = \beta_0 + \beta_1 \ln X_i + u_i$

.....

1.7 LET US SUM UP

In this Unit we discussed the importance and scope of econometrics. Also, we distinguished between theoretical econometrics and applied econometrics. In econometric models we follow certain steps: specification of the model in quantitative terms, collection of data, estimation of the model, testing of hypothesis and interpretation of results, and drawing inferences.

Econometric models are stochastic in nature; we add a stochastic error variable to the regression model. The stochastic error is a random variable. It includes the omitted explanatory variables, error due to wrong specification of the functional form, and selection of wrong sample.

Data that we consider for estimation of parameters take occur through a data generation process (DGP). The DGP is not known to us; we try to conjecture it through some econometric model. We consider several alternative models and opt for the best out of these alternative econometric models.

1.8 KEY WORDS

Alternative Hypothesis	: In hypothesis testing, alternative hypothesis states a condition that is opposite to the null hypothesis. It is expressed as $H_1: \beta_2 \neq 0$, i.e., the slope coefficient is different from zero. It could be positive or negative.
Classical Linear Regression Model	: It refers to a linear regression model that establishes a linear relationship between the variables, based on certain assumptions regarding the error variable.
Confidence Interval	: It is the range of values that determines the probability that the value of the parameter lies within the interval.
Deterministic	: It represents the systematic component of the regression equation. It is the expected value of the

Component	dependent variable for given values of the explanatory variable.
Econometric Model	: These are statistical models specifying relationship between relationships between various economic quantities.
Estimation of Parameters	: This process deals with estimating the values of parameters based on empirical data that has a random component.
Null Hypothesis	: It is the hypothesis that there is no significant difference between specified population, the observed difference is mainly due to sampling or experimental error.
Parameter	: A quantity or statistical measure for a given population that is fixed. The mean and the variance of a population are population parameters.
Population Regression Function (PRF)	: A population regression function hypothesizes a theoretical relationship between a dependent variable and a set of independent or explanatory variables. It is a linear function. The function defines how conditional expectation of a variable Y responds to the changes in independent variable X.
Statistical Inference	: It refers to the process of deducing properties of underlying probability distribution of the parameters by analysing data.
Stochastic Error	: The error variable represents the influence of those variables that are not included in the regression model. It is evident that even if we try to include all the factors that influence the dependent variable, there exists some intrinsic randomness between the two variables.

1.9 SOME USEFUL BOOKS

Maddala, G. S., and Lahiri, K. (2009), *Introduction to Econometrics*, Fourth Edition, John Wiley & Sons.

Verbeek, M. (2017), *A Guide to Modern Econometrics*. Fifth Edition, John Wiley & Sons.

Wooldridge, Jeffrey M. (2022), *Introductory Econometrics: A Modern Approach*, Cengage Learning.

1.10 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Econometric model is a simplification of reality. It retains the major features while excludes the finer details. It is expressed in mathematical form. It can be estimated by using real life data.
- 2) Refer to Sub-Section 1.2.1 and answer.
- 3) Refer to Sub-Section 1.2.3 and answer.

Check Your Progress 2

- 1) The error variable signifies the stochastic nature of relationship. The error term includes the omitted explanatory variables, error due to wrong specification of the functional form, and error due to selection of wrong sample.
- 2) Data generation process reflects the reality. It indicates the process through which values of different variables pertaining to an observation occur. The DGP is not known to us. We try to approximate it with certain econometric model.
- 3) (i) A unit change in X results in β_1 units of change in Y .
(ii) A unit change in results in β_1 per cent of change in Y .
(iii) A unit per cent change in X results in β_1 per cent of change in Y .

UNIT 2 REVIEW OF STATISTICAL CONCEPTS*

Structure

- 2.0 Objectives
- 2.1 Introduction
- 2.2 Statistical Inference
 - 2.2.1 Point Estimation
 - 2.2.2 Interval Estimation
 - 2.2.3 Properties of a Good Estimator
- 2.3 Asymptotic Properties of an Estimator
 - 2.3.1 Law of Large Numbers
 - 2.3.2 Central Limit Theorem
- 2.4 Hypothesis Testing
 - 2.5.1 Null Hypothesis and Alternative Hypothesis
 - 2.5.2 Type-1 Error and Type-2 Error
 - 2.5.3 Level of Significance
 - 2.5.4 Critical Region
 - 2.5.5 Confidence Interval
- 2.5 Estimation Methods
 - 2.5.1 Method of Least Squares
 - 2.5.2 Method of Maximum Likelihood
 - 2.5.3 Method of Moments
- 2.6 Let Us Sum Up
- 2.7 Key Words
- 2.8 Some Useful Books
- 2.9 Answers/Hints to Check Your Progress Exercises

2.0 OBJECTIVES

After going through this unit, you will be able to

- differentiate between a parameter and a statistic;
- distinguish between a ‘point estimate’ and an ‘interval estimate’;
- discuss the properties of a ‘good estimator’;

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- write a note on the asymptotic properties of an estimator;
- define the concepts of ‘weak and strong law of large numbers’;
- state the ‘Central Limit Theorem’ illustrating the same with an example;
- explain the different terminologies used in the context of the testing of a ‘statistical hypothesis’;
- apply the least squares method to estimate the parameters of a linear regression model;
- elucidate the method of ‘maximum likelihood estimation (MLE)’;
- derive the ‘maximum likelihood estimates’ of the parameters of standard distributions like Binomial, Poisson and Normal;
- list the properties of MLE; and
- describe the ‘method of moments’ estimation with illustrations.

2.1 INTRODUCTION

Estimation and hypothesis testing are the two parts of classical statistical inference that is employed all through in econometrics. The process of estimating a population parameter by a random sample drawn appropriately from a population is referred to as ‘estimation’. The process of assessing whether to accept or reject a hypothesis, making an assumption about the value of a population parameter based on sample information, is done in ‘hypothesis testing’. Inferences (or conclusions) about the relationship between a dependent variable and independent variable, in probabilistic terms, is done in hypothesis testing of regression models (or equations) fitted from sample data.

The aggregate of objects under consideration in any statistical research is referred to as the ‘population’. In sampling theory, population refers to the ‘target group’ from which the samples are drawn. A sample is a finite portion of the population selected with the objective of exploring its properties. The sample size refers to the number of units in the sample. The statistical constants of the population like mean (μ), variance (σ^2), skewness (β_1), Kurtosis (β_2), moments (μ) correlation coefficient (ρ) etc are known as ‘parameters’. We compute corresponding statistical constants for the sample drawn from the population. Such statistical constants of the sample such as mean ($\bar{x}$), variance (s^2), skewness (b_1), Kurtosis (b_2), moments (m_r), and correlation coefficient (r) are referred to as ‘statistics’.

2.2 STATISTICAL INFERENCE

The purpose of sampling is to investigate a characteristic of the population based on sample observation. A carefully selected sample is supposed to reveal their characteristics. Thus, in statistical inference, we deduce information about the population (more precisely about the population parameters) based on an analysis

of the sample. This process is known as ‘statistical inference’. To begin with, we may have no information about the population in terms of their parametric values. Therefore, estimation is required. Secondly, some data or hypothetical values for the parameter may be available, and it may be required to test whether a hypothesis formulated is tenable. This brings us to the need for ‘hypothesis testing’. The estimation may be of two types, viz., (i) point estimation, and (ii) interval estimation.

2.2.1 Point Estimation

A point estimate of a parameter is a specific value of a ‘statistic’. The term ‘statistic’ here refers to the aggregate value of sample observations (e.g. mean, standard deviation). Thus, the term ‘statistic’, more generally refers to a function of sample observations. The value obtained by computing the statistic (e.g. $\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$), for a particular sample is called the ‘estimate’. Thus, if $(x_1, x_2, \dots, x_n)$ is a random sample from the population and the statistic is more generally denoted by ‘ T ’, then

$$T = f(x_1, x_2, \dots, x_n) \quad \dots (2.1)$$

is the computed value of T using the sample data $x_1, x_2, \dots, x_n$. In other words, T is the ‘estimator’ and the value taken by it for the sample observations, say t , is an ‘estimate’. An estimator is thus a general rule or a formula for estimating an unknown population parameter. Thus estimator (i.e., the formula) will be the same for multiple samples. The estimate (i.e., the value obtained by applying the formula) will vary from sample to sample. Any particular estimate obtained from a sample is a ‘point estimate’. The distribution of many such point estimates, obtained from different samples, is called the ‘sampling distribution’ of the ‘statistic’.

2.2.2 Interval Estimation

Let θ be a population parameter and let T_1 and T_2 be two functions based on sample observations such that $P(T_1 \leq \theta \leq T_2) = (1 - \alpha)$. Here, α is the level of significance (usually taken as a low value of 1 or 5 percent). Now, for the parameter θ , with the confidence coefficient as ‘ $1 - \alpha$ ’, the interval (T_1, T_2) is termed as an ‘interval estimate’ or a confidence interval. The statistics T_1 and T_2 are called the lower and upper confidence limits for the parameter θ . More specifically, to illustrate, if we draw a random sample (X) from a population whose parameter values (mean, variance) are not known, and the sample observations are 2, 3, 5, 6 & 9, then the sample mean which is $\bar{X} = 5$ represents the point estimate of the unknown population mean. However, if we say the true population mean lies between $(\bar{X} - 0.5)$ and $(\bar{X} + 0.5)$, i.e., between 4.5 and 5.5, then the range of values from 4.5 to 5.5 becomes an ‘interval estimate’ of the ‘population mean’.

2.2.3 Properties of a Good Estimator

There are four criteria for a good estimator. These are: (i) unbiasedness, ii) consistency, iii) efficiency and iv) sufficiency. Let us formally define each one of this now.

Unbiasedness: A statistic ‘ T ’ is said to be an unbiased estimator of a parameter θ , if the expected value of T is θ , i.e.,

$$E(T) = \theta \quad \dots (2.2)$$

As an illustration of this property, we can see how the sample mean $\bar{x}$ is an unbiased estimator of the population mean μ . We have: sample mean: $\bar{x} = \frac{x_1+x_2+\dots+x_n}{n}$ and the population mean (μ) = $\frac{X_1+X_2+\dots+X_n}{N}$. We need to verify that $E(\bar{x}) = \mu$. Each of the sample member $x_1, x_2, \dots, x_n$ is a random variable because their values in any particular sample depends on its probability of being included in the sample. For instance, in SRSWR any of the population members $X_1, X_2, \dots, X_n$ may appear at the ‘ i^{th} ’ drawing, i.e., x is the random variable with the probability distribution as follows:

X_i	X_1	X_2		X_n
Probability	$\frac{1}{N}$	$\frac{1}{N}$		$\frac{1}{N}$

$$E(x_i) = X_1 \cdot \frac{1}{N} + X_2 \cdot \frac{1}{N} + \dots + X_n \cdot \frac{1}{N}$$

$$= \frac{1}{N} (X_1 + X_2 + \dots + X_n)$$

$$= \mu = \text{population mean}$$

$$E(\bar{x}) = E\left[\frac{1}{n} (X_1 + X_2 + \dots + X_n)\right]$$

$$= \frac{1}{n} [E(X_1) + E(X_2) + \dots + E(X_n)]$$

$$= \frac{1}{n} [\mu + \mu + \dots + \mu] = \frac{1}{n} n\mu = \mu$$

Minimum Variance Unbiased Estimator (MVUE): Since the sample drawn can be many, and each sample has its own variance, this needs us to consider the term ‘minimum variance unbiased estimator’. The minimum variance unbiased estimator will have the smallest variance in the class of all unbiased estimators. In other words, if T_m is the minimum variance unbiased estimator of any population parameter θ , then $E(T_m) = \theta$ and $\text{Var}(T_m) < \text{Var}(T)$, for any T with $E(T) = \theta$.

Consistency: Let $X_1, X_2, \dots, X_n$ be the random variables drawn from a population with parameter θ . Let ‘ T ’ be a statistic computed as an estimate of the population parameter θ . The statistic ‘ T ’ is said to be a consistent estimator of the parameter θ , if:

$$\lim_{n \rightarrow \infty} P(T \rightarrow \emptyset) = 1 \quad \dots (2.3)$$

Note that for any distribution, sample mean ($\bar{X}$), is a consistent estimator of the population mean (μ) and sample variance S^2 is a consistent estimator of the population variance σ^2 .

Efficiency: The most efficient estimator is one which has minimum variance among all consistent estimators. If T_1 and T_2 are consistent estimators of a parameter $\emptyset$, such that

$$\text{Var}(T_1) < \text{Var}(T_2), \forall n$$

then T_1 is said to be more efficient than T_2 . If T_1 is the most efficient estimator with variance V_1 and T_2 is the other estimator with variance V_2 for a parameter $\emptyset$, then the efficiency of the estimator is defined as:

$$\text{Efficiency} = \frac{V_1}{V_2}$$

Since $V_1 \leq V_2$, the efficiency of any estimator cannot exceed unity.

Sufficiency: Let $X_1, X_2, \dots, X_n$ be the random variables of size n drawn from a population with parameter $\emptyset$. Let ' T ' be a statistic computed to estimate the parameter $\emptyset$. If ' T ' contains complete information in the sample regarding the parameter $\emptyset$, then it is called a sufficient estimator. Now, if a statistic ' T ' is computed on the basis of a random sample of size n , from a population with probability density function (p.d.f) $f(X, \emptyset)$, then T is a sufficient estimator of $\emptyset$ if the conditional distribution of $X_1, X_2, \dots, X_n$ for a given value of T is independent of $\emptyset$. This means, the conditional probability $P[X_1 \cap X_2 \cap \dots \cap X_n | T = t]$ does not depend on (or contain) $\emptyset$.

Mean Squared Error: When comparing two estimators, a need to make a choice between the two may arise. Suppose we have two estimators, one of which has lower bias but more variance than the other. By this, we mean:

$$(\text{bias } T_1) > (\text{bias } T_2) \text{ but } \text{Var}(T_1) < \text{Var}(T_2).$$

Now, the mean-squared error for T_1 is defined as:

$$\text{MSE}(T_1) = E[T_1 - \emptyset]^2 = \text{Var}(T_1) + (\text{bias } T_1)^2$$

Now, if $\text{MSE}(T_1) < \text{MSE}(T_2)$, we say that T_1 has a lower mean-squared error and accept it as an estimator of $\emptyset$ based on the mean-squared error property.

2.3 ASYMPTOTIC PROPERTIES OF AN ESTIMATOR

The asymptotic properties relate to an estimator's distribution as the sample size grows larger and approaches infinity. The following are the two asymptotic properties of an estimator.

Asymptotic Unbiasedness: An estimator T is an asymptotically unbiased estimator of θ if: $\lim_{n \rightarrow \infty} E(T) = \theta$. It means, as the sample size approaches infinity, the estimator T , which was previously biased, becomes unbiased.

Consistency: The asymptotic property of consistency relates to the behaviour of an estimator's bias and variance as the sample size approaches infinity. If increasing the sample size reduces the estimator's bias and variance, with the trend continuing till both the bias and variance reaches zero as $n \rightarrow \infty$, the estimator is said to be consistent. Thus, for T to be a consistent estimator, it must satisfy the following two conditions: (i) $\lim_{n \rightarrow \infty} [E(T) - \theta] = 0$, and (ii) $\lim_{n \rightarrow \infty} Var(T) = 0$.

2.3.1 Law of Large Numbers

Let $X_1, X_2, \dots, X_n$ be a sequence of i.i.d. (independent and identically distributed) random variables each with mean μ and standard deviation σ . Let the sample mean be given by: $A_n = \frac{X_1 + X_2 + \dots + X_n}{n}$. Now, the weak law of large numbers states that for all $\alpha > 0$: $\lim_{n \rightarrow \infty} P(|A_n - \mu| > \alpha) = 0$ [or $\lim_{n \rightarrow \infty} P(|A_n - \mu| < \alpha) = 1$]. The strong law of large numbers, on the other hand, requires that $\lim_{n \rightarrow \infty} P(A_n = \mu) = 1$ [or $\lim_{n \rightarrow \infty} P(|A_n - \mu| = 0) = 1$].

The concept of sampling distribution of a statistic can now be revisited as follows. If $x_1, x_2, \dots, x_n$ are random variables of same sample size n drawn from a population with their respective means as $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n$, the sampling distribution of these means is the 'probability distribution of these sample means'. If we further assume that the samples are drawn from a normal population, with the n samples drawn are 'independent and identically distributed' (i.i.d.), it leads us to a result called the 'central limit theorem' discussed next.

2.3.2 Central Limit Theorem

Let $X_1, X_2, \dots, X_n$ be n 'independent and identically distributed' (i.i.d.) random variables, each of size n , and drawn from a population having mean μ and variance σ^2 , then as the sample size n becomes large, the distribution of the sample mean $\bar{X}$ approaches normal distribution with mean μ and variance $\frac{\sigma^2}{n}$. In other words, we have: $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ as $n \rightarrow \infty$. You are familiar with the concept of standardised normal variable. It is obtained by subtracting the mean from the variable and dividing it by the standard deviation such that

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \quad \dots (2.4)$$

The variable Z as defined in equation (2.4) approaches to the standard normal distribution, i.e., a normal variate with zero mean and unit variance, as n becomes large. Let us consider an illustration now.

We wish to test whether a coin tossed 30 times, with 'Heads' obtained in 13 tosses is unbiased. Here, $n = 30$, and $\bar{x} = \frac{13}{30} = 0.43$. If the coin is unbiased, then p